

Bitcoin Single-Sig Key Ceremony - v0.2

Procedure & Ceremony Record

Purpose & Scope

This document records the procedure for generating a single-signature Bitcoin wallet's seed phrase from dice-rolled entropy, entered directly into an air-gapped hardware wallet, and for backing up, verifying, and test-funding the resulting wallet before it is trusted with real funds. It is a checklist to follow during the ceremony and a record of what was actually done afterward.

This document does not contain any seed words, passphrases, or private keys. It records how and when the wallet was created, not the wallet's secret material itself.

It is used alongside three supporting printed documents:

- ☐ 1-Binary-to-Decimal-Conversion-Table
- ☐ 2-BIP-39-Decimal-Binary-Word-Reference-Table
- ☐ 3-Seed-Phrase-Paper-Backup

Materials Required

- ☐ Pen
- ☐ The three printed reference documents listed above
- ☐ A standalone calculator (never a phone connected to the internet)
- ☐ An air-gapped hardware wallet
- ☐ A microSD card or USB stick, used only to carry a firmware for updates (if necessary) whose signature was verified previously
- ☐ A quiet, private room with no video surveillance; all phones and other cameras left in another room or in a Faraday bag
- ☐ At least 6–10 dice, ideally casino-grade and from more than one manufacturer or style
- ☐ A cup or box to roll the dice fairly
- ☐ A dedicated computer with the Bitcoin software wallet installed (e.g. Sparrow), used only for this purpose
- ☐ Steel backup kit (steel plate, Cryptosteel, or similar) and stamping tool
- ☐ An industrial-grade microSD card for the encrypted backup (if your wallet supports this)
- ☐ A secure password manager for the wallet's non-secret configuration data (descriptor, xpub, fingerprint, derivation path, watch-only wallet login password)
- ☐ A personal and private Bitcoin node

Procedure

Phase 1 - Preparation

- ☐ 1. Gather all materials from the checklist above.
- ☐ 2. Verify the hardware wallet's authenticity per the vendor's official verification process before any entropy touches it and do the first setup of the new wallet.
- ☐ 3. Check the device's current firmware version; if a newer version is available, update it now using the microSD/USB whose firmware signature you verified beforehand.
- ☐ 4. Confirm the environment is offline: the hardware wallet stays air-gapped throughout.
- ☐ 5. Verify the printed 2048-word reference table (2-BIP-39-Decimal-Binary-Word-Reference-Table) is identical to the canonical BIP-39 wordlist on GitHub.

Phase 2 - Rehearsal (test seed, discarded afterward)

- ☐ 1. Roll dice and generate one or two throwaway test seeds using the 1-Binary-to-Decimal-Conversion-Table worksheet, to rehearse the workflow and confirm the device behaves as expected (the same steps as in Phase 3).
- ☐ 2. Export the test wallet's xpub into Sparrow (or your chosen software wallet) and verify the receive addresses shown.
- ☐ 3. Cross-check those addresses, the master fingerprint, change addresses, and zpub against an independent tools like (Ian Coleman's BIP-39 tool and bitcoiner.guide/seed) and/or a second hardware wallet, if available.
- ☐ 4. Discard the test seed and any test wallet data once the rehearsal is confirmed working.

Phase 3 - Real Entropy Generation

- ☐ 1. Roll the dice, filling in the worksheet row by row until words 1–23 (and, optionally, row 24) are complete. The number of rolls needed varies, some rolls produce 2 bits, some produce 1, so it isn't a fixed roll count like before.
- ☐ 2. Convert each roll using the key printed on the worksheet (☐=00, ☐=01, ☐=10, ☐=11, ☐=0, ☐=1), filling in the boxes left to right. If a roll's bits don't all fit in the row's remaining boxes, write what fits and carry the leftover bit into the first box of the next row.

Example: rolling: ☐ ☐ ☐ ☐ ☐ ☐ ☐ → "01 1 11 0 10 00 0"

- ☐ 3. Calculate each row's decimal value using the place-value columns, using the calculator to sum the values.

☐	1	1	1	1	☐	1	☐	☐	☐	☐	# 976	Word 1: key
1024	512	256	128	64	32	16	8	4	2	1		

- ☐ 4. Look up each decimal value in the verified reference table to get the corresponding words. Once you have words 1–23, write them into the BIP-39 Seed Phrase Paper Backup (3-Seed-Phrase-Paper-Backup). If your wallet supports manual entry for word 24, also roll the 3 extra entropy bits on row 24, otherwise the wallet will generate word 24 entirely on its own.

Phase 4 - Enter Into the Device

- ☐ 1. Power on the air-gapped wallet and select 'Import an existing seed' (or whatever your device calls the option that lets you enter your own seed words).
- ☐ 2. Enter the dice-derived words exactly as recorded.
- ☐ 3. If you rolled the 3 extra entropy bits for word 24 and your wallet supports manual entry, enter that partial word as well; otherwise let the device generate word 24 entirely on its own. Add the completed 24th word to the BIP-39 Seed Phrase Paper Backup (3-Seed-Phrase-Paper-Backup).
- ☐ 4. Confirm the device reports the checksum as valid.

Phase 5 - Record the Backup

- ☐ 1. Fill in the paper backup form: the word grid, master fingerprint, derivation path, script type, notes on seed generation, and the SeedQR if your wallet allows it.
- ☐ 2. Read every word back from the device screen (if it lets you do so) and compare it word-by-word against the paper.
- ☐ 3. Confirm the master fingerprint on the form matches the device's display.
- ☐ 4. Export the wallet into Sparrow (watch-only) and verify the addresses shown there match the device.
- ☐ 5. Record the wallet's non-secret configuration data: xpub, zpub, master fingerprint, derivation path, and the Sparrow watch-only wallet login, into your password manager (e.g. Bitwarden). Never store the seed words or passphrase there.

Phase 6 - Transfer to Steel

- ☐ 1. Stamp or engrave the words onto the steel backup.
- ☐ 2. Read the steel backup back, word-by-word, against the paper to catch any stamping error.
- ☐ 3. If the device supports an encrypted microSD backup, create it now.

Phase 7 - Small-Amount Funding Test + Dry-Run Recovery

- ☐ 1. Send a small, affordable-to-lose test amount to the wallet's first receive address.
- ☐ 2. Confirm it arrives correctly in Sparrow.
- ☐ 3. Restore the wallet from steel, paper, and microSD.
- ☐ 4. Confirm the restored master fingerprint matches.
- ☐ 5. Confirm the first several receive addresses match exactly.
- ☐ 6. Spend that same test amount back out, to verify everything was done properly.
- ☐ 7. Only after this succeeds, consider the wallet ready for real funds, and store the steel backup and the other two copies (microSD and paper) in secure, separate locations.

Ceremony Record

Complete after the ceremony. Contains no seed material - safe to store with other administrative records.

Session Details

Date: _____

Start time: _____

End time: _____

Location: _____

Operator: _____

Witness (if any): _____

Wallet Parameters

Derivation path: _____

Script type: _____

Master fingerprint: _____

Hardware wallet model: _____

Firmware version: _____

Verification Confirmed

- ☐ Wordlist verified against canonical source
- ☐ Rehearsal (test seed) completed and cross-checked
- ☐ Checksum reported valid by device
- ☐ All 24 words read back and matched against paper
- ☐ Sparrow addresses matched device addresses
- ☐ Steel backup engraved and read back for accuracy
- ☐ Dry-run recovery from steel succeeded (fingerprint + addresses match)
- ☐ SeedQR tested (if used)
- ☐ MicroSD with backup tested (if used)
- ☐ Small-amount funding test: deposit and respond both succeeded

Notes

Sign-off

Completed by (name / signature): _____

Date completed: _____

Next scheduled backup check-in: _____